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$$\int \cos^5 x dx = \int \cos^4 x \cos x dx$$
$$\int (\cos^2 x)^2 \cos x dx = \int (1 - \sin^2 x)^2 \cos x dx$$
$$t = \sin x \Rightarrow dt = \cos x dx$$
$$\int (1 - t^2)^2 dt = \int (t^4 - 2t^2 + 1) dt$$
$$= \int t^4 dt - 2 \int t^2 dt + \int t^0 dt$$
$$= \frac{t^5}{5} - \frac{2t^3}{3} + t + C$$
$$= \frac{1}{5} \sin^5 x - \frac{2}{3} \sin^3 x + \sin x + C$$

References